UGANDA ADVANCED CERTIFICATE OF EDUCATION (UACE)

P425/1 PRINCIPAL MATHEMATICS (Paper 1)

S.6 END OF TERM ONE 2026

EAGLES WING'S HIGH SCHOOL

Time: 2 Hrs. 45 minutes

INSTRUCTIONS TO CANDIDATES:

1. This paper consists of **three (3)** sections; **A, B** and **C** with a total of **five** examination items.
2. Section **A** has one compulsory item while section **B** and **C** each have two items. Attempt only one item in each section i.e. One item in section B and one item in section C.
3. All in all, a total of **3 items** should be attempted. Any additional item(s) answered will not be given any score(s).
4. Begin each item on a fresh page and clearly number them as they appear in the examination s paper.
5. Silent, non-programmable calculators and mathematical tables are allowed. Where necessar y, graph paper is provided.
6. Tidy handwriting, clear presentation of work and meaningful mathematical judgments or co nclusions will increase your chances of excelling.

FOR SCORER'S USE ONLY

SECTION	A	B		C	
ITEM	1	2	3	4	5
SCORE					
GRADE					
INITIAL					

SECTION A (GEOMETRY)

This item is compulsory

Item 1

Stephen is a well-known agriculturalist in Mbale and the entire Bugisu sub-region. He mainly grows Arabica coffee in his land located at the slopes of Mt. Elgon and as well has a huge herd of cattle. In the recent years, he has been facing a challenge of cattle rustling(theft) and now intends to build a circular kraal that passes through point **P(2,9), Q(5,0)** and **R(−1,8)** but he is struggling to know the Centre and radius of the circular kraal and as well the equation described by the kraal. One evening when he was driving, a straight light ray struck tangentially to his parabolic mirror at point $(4,-1)$ and it was also noted that the parabolic mirror could be represented by the equation $x^2+y^2-30x+6y+109=0$. In order to ease movement in his arabica coffee farm, a straight road, $\mathbf{R_1}$ is to be constructed passing through point **A(2,2)** and point **B(3,3)** while another, $\mathbf{R_2}$ will be constructed passing through **C(2,0)** and **D(−1, −1)** and at the point of intersection of these two roads will be a simple navigation round about yet they need to know the acute angle that one has to turn to access road $\mathbf{R_1}$ if he was originally passing in road $\mathbf{R_2}$ or vice versa. Similarly, there is a new company that is purchasing coffee and they transport and their plane usually tunes to signals at angles θ satisfying the equation $12\cos\theta+5\sin\theta=2$ whose working range is **$-360^0 \leq \theta \leq 360^0$.**

Stephen now faces a big challenge of knowing the radius, Centre and equation of the circular kraal. He is as well interested in knowing the equation of the straight light ray that struck his parabolic mirror, the point to put a simple round about and the acute angle to turn to locate a given line. He also needs to know the angles satisfying the signal equation.

Task

Help Stephen to determine;

a) The radius and Centre of the kraal hence deduce its equation (expand and simplify your response)

b) The equation of the light ray that struck his mirror

c) By mathematical calculations where exactly the simple round about in the coffee farm should be placed and as well determine the acute angle one turns to locate road $\mathbf{R_1}$ if he/she was originally passing in road $\mathbf{R_2}$

d) The values of θ satisfying the signal equation.

SECTION B (ALGEBRA)

Attempt any ONE item in this section

Item 2

At Namanve industrial park, trucks move following a particular road pattern described by the quadratic equation $2x^2 - 3x + 4 = 0$ whose roots are $\propto$ and β which are obtained without solving the equation. However, when the minister of works and transport went for inspection, he realized that there was congestion and opted to construct a new road of the same nature as the existing one but with its roots being $\frac{1-\propto}{\beta}$ and $\frac{1-\beta}{\propto}$.

Later on, the minister also wanted to estimate the value of $\sqrt[3]{24}$ and he had been directed to begin by

Expanding $(1-x)^{\frac{1}{3}}$ up to the term in x^3 and then substitute $x = \frac{1}{9}$ in the result to obtain the $\sqrt[3]{24}$ correct to 4 decimal places. After this inspection, a team of 10 members went for a dinner and had to sit in a round table but they were interested in knowing how many possible ways they could sit in the table. In order to make their party colorful, they had to withdraw some money but the pin was written as **113r** where r is a value satisfying the equation $\log_3 r + \log_9 3r = 2$ and had to look for r to get the actual PIN. During another analysis conducted by the minister of works and transport, a certain road M was found to

be $\frac{2\sqrt{2}}{3-\sqrt{2}} + \frac{4+\sqrt{2}}{3+\sqrt{2}}$ kilometres and he was interested in the exact value in the form $a + b\sqrt{2}$ where a and b are integers.

Task

a) Determine the equation of the new road to be constructed

b) Expand $(1-x)^{\frac{1}{3}}$ up to the term in x^3 hence by substituting $x = \frac{1}{9}$, estimate the value of $\sqrt[3]{24}$ correct to 4 decimal places

c) Determine the actual PIN by first finding the value of **r** in the logarithmic equation

d) What is the actual length of road M in the form $a + b\sqrt{2}$ where **a** and **b** are integers, stat

e the values of **a** and **b**

Item 3

At St. Lawrence SSS Sonde, the security team was tasked to install cameras in all major school prem ises including the library. The technical team realized that a certain plane had an equation $x - y + z = 1$, another had $2x + y - z = 5$ and the last one had $x + 3y + 2z = 5$. They came up with a plan that the camera in the library be installed at the point on intersection of the above planes.

In order to make this plan come to a success, they decided to collect school fees using a geometric p attern. In week 2, they collected **60 million**, in week 4 they collected **240 million**. Their intentions are to use 1% of the total amount collected in the **first 5 weeks** for installing cameras and are interested in knowing how much it is.
The school has also planned that a team of 13 non-teaching staff be chosen to help in the process b ut since they have a total of **20** non-teaching staff, they need to know how many ways they can cho ose the required staff.

When they contacted the head teacher, she added that they were to build **b** bathrooms such that th e value **b** satisfies the equation $25^b - 5^b - 600 = 0$ but now they are confused on how many bathroo ms were to be built. Meanwhile, the head teacher also proposed that the school should also invest in a solar power back up system and upon consulting the school engineer, he modelled the current as $I = \frac{3+4i}{1-2i}$ in amperes.
They need to express this current in the form $a + bi$ and determine its modulus, principal argument a nd its polar form for proper tuning.

Task

Help the school to determine;

a) Where the camera in the library should be placed hence determine how much money is need ed for installing cameras in the school.
b) How many possible ways the 13 non-teaching staff can be chosen from the total of **20** nont eaching staff
c) The number of bathrooms that are to be built as seen by the head teacher.
d) Express the current in the form $a + bi$, determine its modulus and principal argument hence write it in polar form for proper tuning.

SECTION C (CALCULUS)

Attempt any <u>ONE</u> item in this section

Item 4

Mama Jennifer's daily cost of producing x batches of mandazi is given by $p(x) = \frac{3}{4 + 5\cos x}$ and her Accountant explains to her that this cost function can be broken down into simpler parts for better analysis. Motivated by her success in the Mandazi business, Mama Jennifer has started a juice business on the side. She uses conical disposable cups that are popular with her customers. These cups are of height, h**cm** and radius **r cm** which are related by $r = \frac{h}{2}$ While filling the cups, the juice is pumped at a rate of $10\pi cm^3 s^{-1}$

and she wants to use this to analyses the rate at which the height of the juice rises in the cups when the height of the juice is **6cm.** Mama Jennifer having received a lot of customers would like construct a good restaurant on an area of land enclosed by the curve $g(x) = 4x - x^2$ **and** $f(x) = x^2 + 4x + 3$ correct to **3** decimal places

Task

Help Mama Jennifer to determine;

a) The integral of $\sum_{-\frac{\pi}{2}}^{\frac{\pi}{2}} p(x)\, dx$

b) The rate at which the height of the juice rises in the cups when the height of the juice is **6cm**

c) Estimate the area of the land where the restaurant will be.

Item 5

Mr. Kapata is a farmer who has built a new fish pond. He needs to understand several mathematical concepts to manage his pond efficiency. Mr. Kapata's fish population follows a growth pattern modelled by the function $P(t) = e^{2t}\cos t$ where P (t) represents the population (in hundreds of fish) after t months. He needs to know the rate of change the fish population, and hence or otherwise, show that this growth function satisfies the fish model given by. $\frac{d^2p}{dt^2} - 4\frac{dp}{dt} + 5p = 0$

He later analyses the quantity of food (x) to give to his fish and realized it satisfied the model **f(x)** = $\frac{x - 9}{x(x-1)(x+3)}$

Task

Help Mr. Kapata to;

a) Determine the rate of change of the fish population, $\frac{dp}{dt}$ and hence or otherwise, show that the growth function ***P(t)*** satisfies the fish model given by $\frac{d^2p}{dt^2} - 4\frac{dp}{dt} + 5p = 0$

b) By partialize the function ***f(x)*** hence find $\sum_{2}^{3} f(x)\,dx$

THE END

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